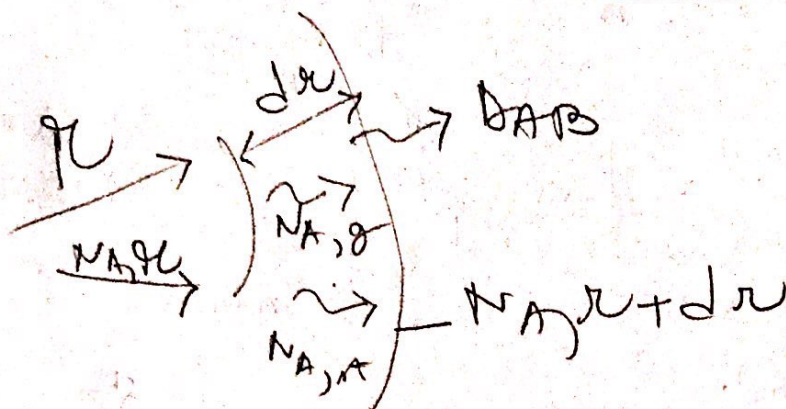


Koku
Ado boe

Chem 312
Spring 2020
HW #13

14-18 > 14.24, 14.33
14.18



Assumption
- Steady state
- 1-D radial diffusion

$$N_{A,r} + \dot{N}_{A,g} - N_{A,r+dr} = \dot{N}_{A,st} \quad (1)$$

$$N_{A,r} = (2\pi r \times 4) \dot{N}_{A,r}$$

$$= 2\pi r D_{AB} \frac{\partial C_A}{\partial r}$$

$$N_{A,r+dr} = N_{A,r} + \frac{\partial N_{A,r}}{\partial r} dr$$

$$= -2\pi r D_{AB} \frac{\partial C_A}{\partial r} - r D_{AB} \frac{\partial}{\partial r} \left[r \frac{\partial C_A}{\partial r} \right] dr$$

$$\dot{N}_{A,g} = -N_A (2\pi r \times 4 dr)$$

$$\dot{N}_{A,st} = \frac{\partial C_A (2\pi r dr \times 4)}{\partial t}$$

Substituting in eq (1):

$$-2\pi r D_{AB} \frac{\partial C_A}{\partial r} - \dot{N}_A (2\pi r dr) + 2\pi r D_{AB} \frac{\partial}{\partial r} \left[r \frac{\partial C_A}{\partial r} \right] dr$$

$$= 2\pi r \frac{\partial r}{\partial t} \frac{\partial C_A}{\partial t}$$

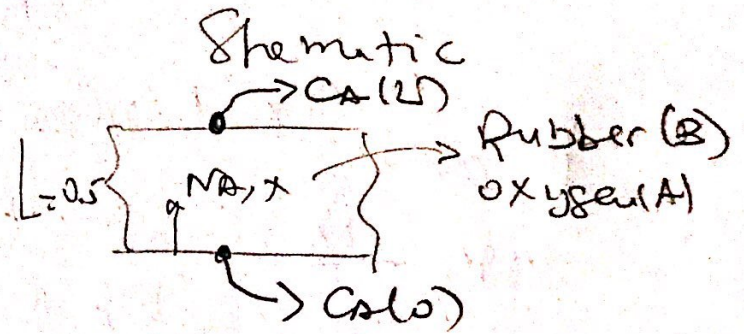
$$-N_A(2\pi r dr) = 2\pi r dr \frac{\partial C_A}{\partial t} - 2\pi r \frac{\partial}{\partial r} \left[r D_{AB} \frac{\partial C_A}{\partial r} \right]$$

$$N_A = \frac{D_{AB}}{r} \cdot \frac{\partial}{\partial r} \left[r \frac{\partial C_A}{\partial r} \right] - \frac{\partial C_A}{\partial t}$$

14.24

Assumptions:

- Steady state
- 1D diffusion
- Stationary medium of uniform total concentration
- perfect gas behavior



We know

$$D_{AB} = 0.21 \times 10^{-9} \text{ m}^2/\text{s}$$

$$T = 25^\circ\text{C} = 298\text{K}$$

$$C_A = 3.12 \times 10^{-3} \frac{\text{kmol}}{\text{m}^3}$$

$$N_{A,x} = D_{AB} \frac{dC_A}{dx} = D_{AB} \frac{(C_{A,w}) - C_A(L)}{L}$$

$$C_{A,w} = 2C_A = 3.12 \times 10^{-3} \frac{\text{kmol}}{\text{m}^3} \times 2 = 6.24 \times 10^{-3} \frac{\text{kmol}}{\text{m}^3}$$

$$N_{A,x} = 0.21 \times 10^{-9} \frac{\text{m}^2}{\text{s}} \left(\frac{6.24 \times 10^{-3}}{0.5 \times 10^{-3} \text{ m}} \right) \text{ kmol m}^{-3}$$

$$N_{A,x} = 1.31 \times 10^{-9} \text{ kmol / s.m}^2$$

$$C_{A,1} = \frac{P_{A,1}}{RT} = \frac{1.5 \text{ bar}}{(0.08314 \frac{\text{m}^3 \text{ bar}}{\text{kmol} \cdot \text{K}})(298 \text{ K})}$$

$$C_{A,1} = 0.0605431 \frac{\text{kmol}}{\text{m}^3}$$

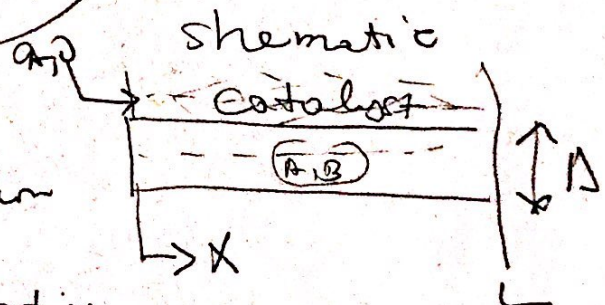
$$C_{A,2} = \frac{P_{A,2}}{RT} = \frac{0.5}{(0.08314 \frac{\text{m}^3 \text{ bar}}{\text{kmol} \cdot \text{K}})(298 \text{ K})}$$

$$C_{A,2} = 0.02018 \frac{\text{kmol}}{\text{m}^3}$$

14.33

Assumptions

- Steady state
- 1-D Diffusion in x direction
- Stationary medium
- Uniform molar concentration
- Stationary medium.



(a)

we know:

$$N_{A,x} - K_1' C_A (\pi D) dx - N_{A,x+dx} = 0$$

$$N_{A,x+dx} = N_{A,x} + (dN_{A,x}/dx) dx$$

$$N_{A,x} = \left(-C D_{AB} \frac{dC_A}{dx} \right) \frac{\pi D^2}{4} = -\frac{\pi D^2}{4} D_{AB} \frac{dC_A}{dx}$$

$$-\frac{dN_A}{dx} dx - K_1' C_A (\pi D) dx = \frac{\pi D^2}{4} D_{AB} \frac{d^2 C_A}{dx^2} - K_1' C_A (\pi D) dx = 0$$

$$\frac{d^2 C_A}{dx^2} - \frac{4 K_1'}{D D_{AB}} C_A = 0$$

⑥ Applying the boundary condition

$$C_A(0) = C_{A0} \quad -D_{AB} \left(\frac{dC_A}{dx} \right)_{x=L} = K_1' C_A(L)$$

$$C_A \rightarrow 0 = T - T_\infty \quad m = \left(\frac{4K_1''}{D_{AB}} \right)^{\frac{1}{2}} \leftrightarrow \left(\frac{4h}{DK} \right)^{\frac{1}{2}}$$

$$D_{AB} \leftrightarrow K$$

$$K_1'' \leftrightarrow h$$

$$C_A(x) = \frac{C_{A0} \cosh m(L-x) + (K_1''/m D_{AB}) \sinh m(L-x)}{C_{A0} \cosh mL + (K_1''/m D_{AB}) \sinh mL}$$

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